control flow diagram

Exercise 1

With space (only for under standings)

```
1.  void main() {
2.     int i, j, k;
3.     readln(i, j, k);
4.
5.     if ((i < j) || (i > k)) {
6.        writeln("then part");
7.
8.        if (j < k)
9.           writeln("j less than k");
10.       else
11.          writeln("j not less than k");
12.    } else {
13.       writeln("else Part");
14.    }
15. }
```

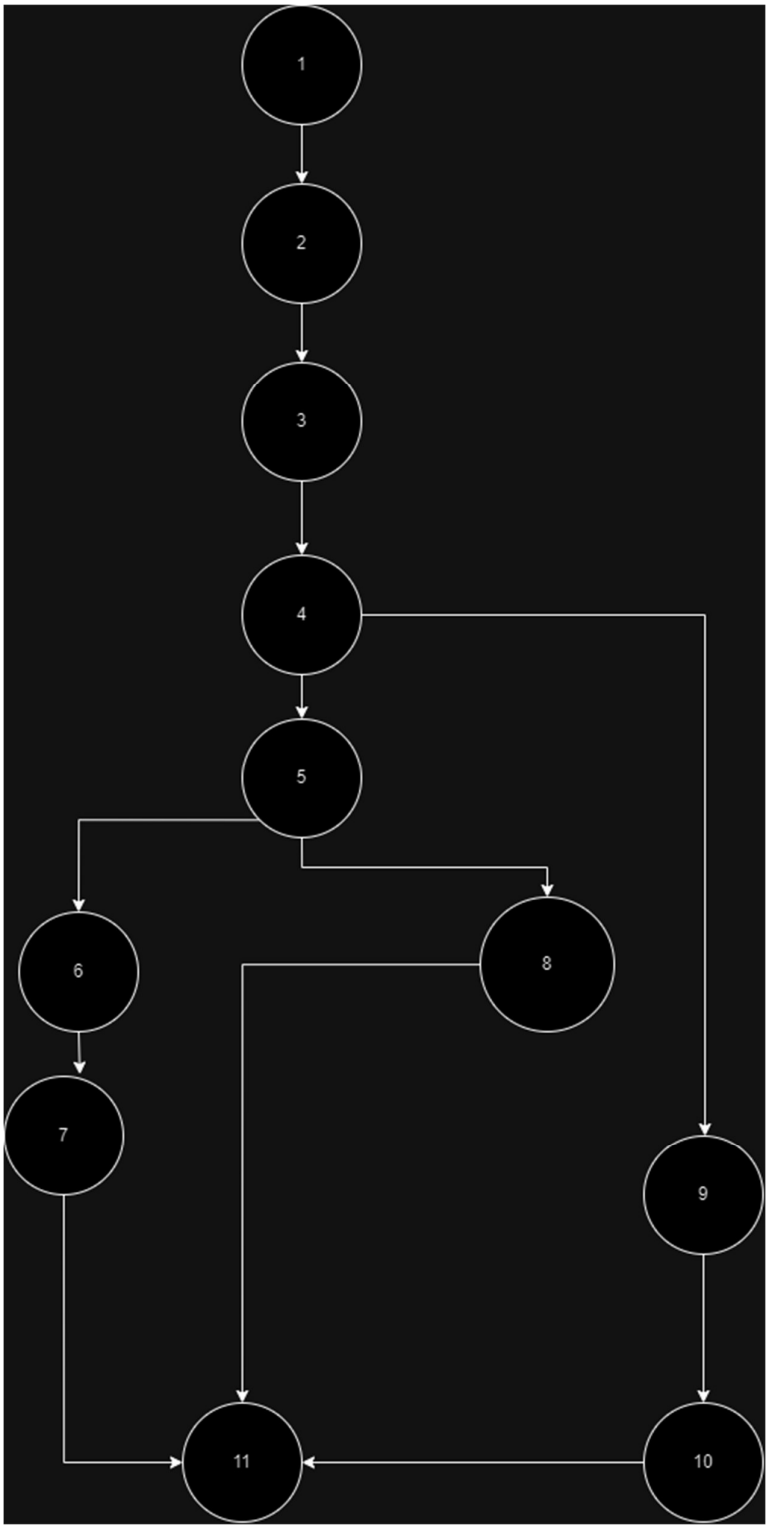
Without space (for use in graph creation)

```
1.void main(){
2.int i,j,k;
3.readln(i,j,k);
4.if((i<j)||(i>k)){
5.writeln("thenpart");
6.if(j<k)
7.writeln("jlessthank");
```

8.else writeln("jnotlessthank");

9.}else writeln("elsePart");

10.}



Number of **Edges** in given flow graph is **12**

Number of **Node's** in given flow graph is **11**

Number of Node that have **exit points** is **1**

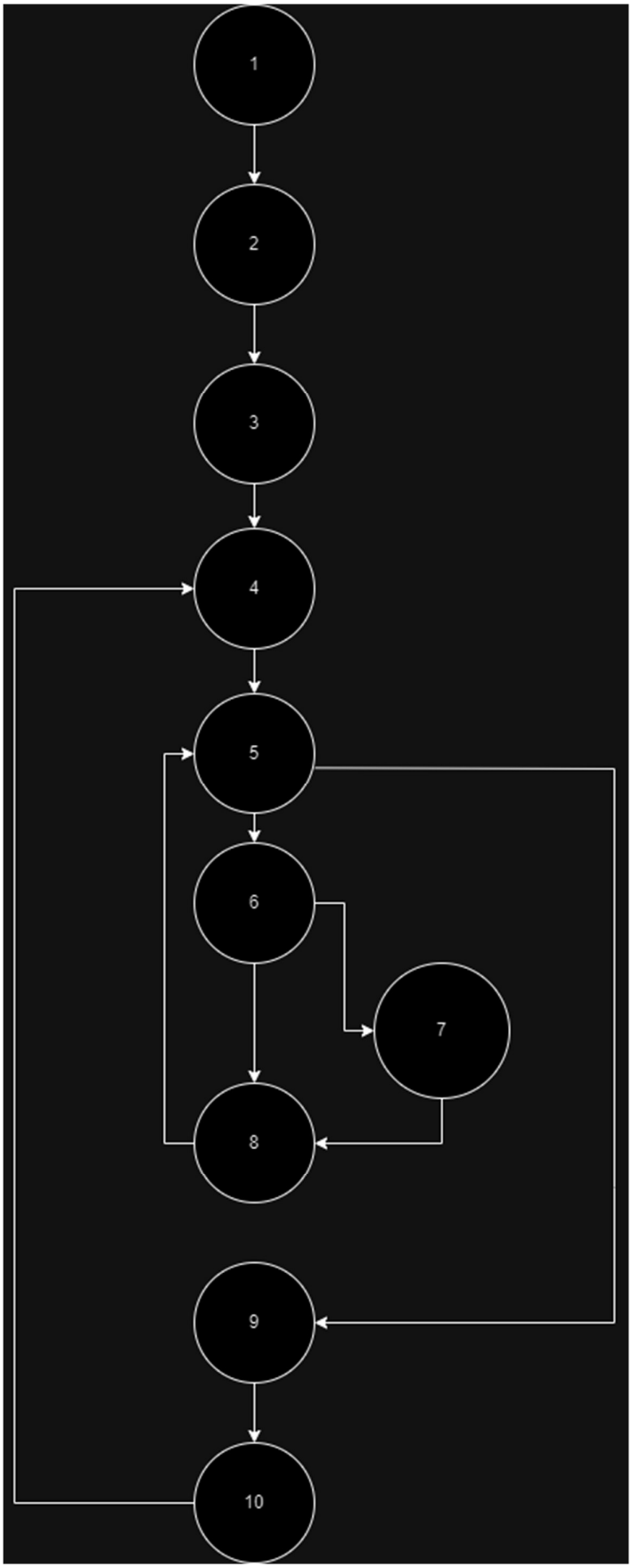
=E-N+2*P

=12-11+2(1)

=3

Example 2

```
i = 0;
n = 4; // N-Number of nodes present in the graph
while (i < n - 1) do
    j = i + 1;
    while (j < n) do
        if A[i] < A[j] then
            swap(A[i], A[j]);
        end do;
        i = i + 1;
    end do;
```


1
2
3
4
5
6
7
8
9
10

Number of **Edges** in given flow graph is **12**

Number of **Node's** in given flow graph is **10**

Number of Node that have **exit points** is **1**

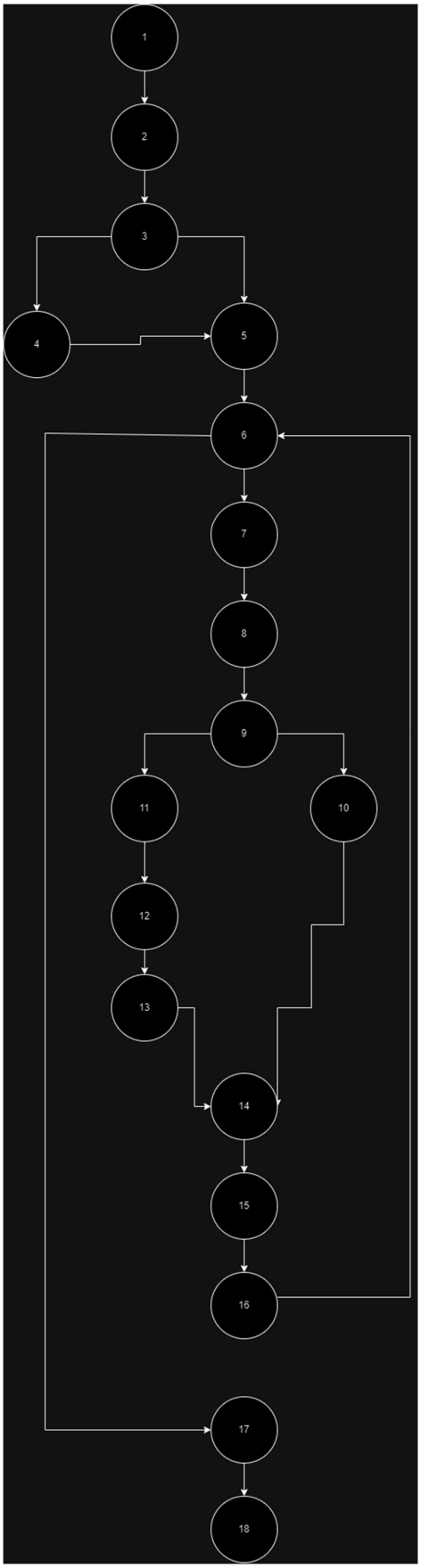
=E-N+2*P

=12-10+2(1)

=4

Example 3

```
public Hashtable countAlphabet(String aString) {
  Hashtable table = new Hashtable();
  if (aString.length() > 4000)
    return table;
  StringBuffer buffer = new StringBuffer(aString);
  while (buffer.length() > 0) {
    String firstChar = buffer.substring(0, 1);
    Integer count = (Integer) table.get(firstChar);
    if (count == null) {
        count = new Integer(1);
     } else {
        count = new Integer(count.intValue() + 1);
     }
     table.put(firstChar, count);
     buffer.delete(0, 1);
  }
  return table;
}
```


1
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16
17
18

Number of **Edges** in given flow graph is **20**

Number of **Node's** in given flow graph is **18**

Number of Node that have **exit points** is **1**

=E-N+2*P

=20-18+2(1)

=4

Example 4